

Blessed Madukoma

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Research Software Engineer

Researching and engineering reliable, observable production AI and data systems

Summary

Systems-focused software engineer and researcher building reliable, observable production AI and data systems across scalable backends, real-time pipelines, and analytics infrastructure, with research at Carnegie Mellon on evaluating AI systems as they evolve post-deployment.

Professional Experience

- Graduate Research Associate** June 2026 – Present
Carnegie Mellon University, Languages and Spatial Technologies Lab Kigali, Rwanda
 - Conducting research on the software engineering of evolving production AI systems, focusing on behavioural regression, statistical verification, and reliable AI evaluation.
 - Performing systematic literature reviews, analysing documented production AI failures, and formalising behavioural descriptions for confidence-bounded regression judgement.
- Graduate Research Assistant** Sep 2025 – May 2026
Carnegie Mellon University, Languages and Spatial Technologies Lab Kigali, Rwanda
 - Built anomaly detection platform & dashboard (FastAPI, PostgreSQL, Redis) over 1.5M+ athletics performances (19k+ events, 2010–2025), supporting anti-doping prioritisation via benchmarked method comparison, ensemble evaluation, and validation against Athletics Integrity Unit sanctions data. [Link](#).
 - Implemented 14 detection methods (statistical, ML, and Bayesian) with explainable consensus flagging, elite-filtered thresholds, and end-to-end audit trails; published at IEEE SDS 2026 (ArXiv: 2604.21953).
- Data Engineering & Analytics Engineer (Contract)** May 2025 – Sep 2025
Ministry of Infrastructure (MININFRA) Kigali, Rwanda
 - Architected Rwanda's first [infrastructure intelligence platform](#), consolidating siloed construction permit and land-use datasets into a centralized system enabling evidence-based urban planning.
 - Built end-to-end data pipelines for construction permits and land-use records, establishing 12+ urban development indicators including permit throughput, and capital allocation trends.
- Software Engineer (MSc Internship)** Jun 2025 – Sep 2025
Huza HR Kigali, Rwanda
 - Built email notification system with multi-stage recruitment templates, automating candidate communications, recruiter alerts, and internal notifications across the hiring pipeline, enabling the company's first fully digital ATS workflow.
 - Implemented core recruitment workflows using Next.js and Node.js, conducting 5+ peer code reviews to establish scalable architecture that onboarded 50+ job seekers and 10+ recruiters within first deployment.
- Software Engineer** Sep 2024 – Dec 2024
The UPANZI Network Kigali, Rwanda
 - Migrated African [datasets hosting platform](#) to Tailwind CSS, reducing bundle size by 30%, improving customization by 25%, and ensuring data privacy standards while maintaining sensitive datasets.
 - Collaborated with cross-functional teams using Agile methods and Git-based version control system, fostering a 15% increase in team productivity and ensuring timely project delivery.
- Golang Backend Engineer** Mar 2024 – Aug 2024
Platnova Remote
 - Implemented CRON-driven vault settlement automation, reducing payment processing time 20% and increasing transaction speed 40%.
 - Revamped transaction receipt system, increasing customer satisfaction 30% and reducing support inquiries 15%.
 - Executed lifestyle-stays package launch, driving customer engagement and increasing revenue by 15%.
- Software Engineer (National Youth Service)** Dec 2022 – Nov 2023
21st Century Technologies Lagos, Nigeria

- Implemented CI/CD pipelines improving deployment efficiency by 70%, reducing release times by 40%, and saving operational costs by 40%.
- Engineered data pipelines ensuring 99.99% uptime via DNS restoration across Tier III/IV data centers with failover protocols.
- Delivered reporting system with visualization dashboards improving management decision-making and customer satisfaction by 45%.

Education

• Carnegie Mellon University

MSc. Information Technology — Software Engineering, Big Data & Applied ML

- **Thesis Focus:** Performance anomaly detection in sports using ML models and statistical methods with systems engineering [application](#). *Advisor: Prof. Prasenjit Mitra.*

• Babcock University

BSc. Software Engineering — First Class Honors

Technical Projects

• Sedge [\[Live\]](#)

Tools: Golang, PostgreSQL, Redis

- Responsible for end-to-end architecture, backend/data pipelines, and Railway-based production deployment for the Sedge ecosystem.
- Building core features – Spark (natural language + no-code analytics/SQL platform) and Architect (drag-and-drop pipeline builder replacing lengthy engineering workflows).

• Performance Anomaly Detection in Athletics [\[Live\]](#)

Carnegie Mellon University | 2025

Tools: Python (FastAPI), PostgreSQL + DuckDB, Redis, HTMX

- Engineered scalable platform processing 1.5M+ global athletics performances (19,000+ competitions, 2010–2025), enabling integrity monitoring complementing expensive biological testing programs.
- Delivered investigator-ready dashboard with 10 competing detection methods (statistical, ML, Bayesian), explainable flagging, and real-world validation against Athletics Integrity Unit sanctions data. Full investigator dashboard: anomaly drill-down, athlete profiles, audit trails, and prioritisation queues.

Technical Skills & Certifications

- **Languages & Core:** Go, Python, TypeScript, JavaScript, SQL
- **Backend & Data Systems:** FastAPI, Node/Express, PostgreSQL, Redis, ETL, Data Modeling
- **ML & Analytics:** Scikit-learn, Time Series Analysis, Statistical Modeling
- **Cloud & DevOps:** AWS, Docker, Git/GitHub, Bitbucket, CI/CD (GitHub Actions)
- **Architecture:** Distributed Systems, Microservices, Event-Driven Architecture
- **Frontend:** Vue.js, Nuxt
- **Certifications:** [Data Engineering Essentials](#), [Data Engineering Introduction](#), [Docker](#), [Agile with Jira](#)

Publications

- [1] Madukoma, B., Mitra, P. (2026). **Performance Anomaly Detection in Athletics: A Benchmarking System with Visual Analytics**. *IEEE Swiss Conference on Data Science and AI (SDS2026)*, Poster Presentation. Zurich, Switzerland. [ArXiv](#).
- [2] Wafula, S., Madukoma, B. (2025). **Unmasking COVID-19 Vulnerability in Nigeria: Mapping Risks Beyond Urban Hotspots**. *Women in Machine Learning (WiML) Workshop @ NeurIPS 2025*. [OpenReview](#).

Relevant Coursework

- **Machine Learning in Production (Prof. Christian Kästner):** Production AI systems, evaluation, MLOps, and responsible AI.
- **Data-Intensive Scalable Systems (Paulo Merson):** Distributed system design, streaming systems, containers, and observability.

Leadership & Achievements

- **Chairman** — Academic Innovation Sub-Committee, Research Committee, CMU Africa (2024–2026).
- **Executive President** — Computer Club, Babcock University (2022). Led 2k+ tech member community.
- **Volunteer** — Google DevFest Kigali (Dec 2024).
- **Participant** — Digital Humanism Summer School, CMU-Africa (2025). [Certificate](#).